Section 5.2 - The Second Derivative

Definition: If f is a differentiable function, then its derivative f' is also a function, so f' may have a derivative of its own, denoted by $(f')' = f''$. This new function f'' is called the **second derivative** of f because it is the derivative of the derivative of f.

$$f''(x) = y'' = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

Example 1: Find the first and second derivative of the following functions:

a) $f(x) = 2x^3 - 14x^2 + 3x - 16$

$f'(x) = 6x^2 - 28x + 3$ ← 1st deriv.

$f''(x) = 12x - 28$ ← 2nd deriv.

b) $y = 3x^2 \ln x$

$y' = (3x^{\cancel{2}})\left(\frac{1}{\cancel{x}}\right) + \ln x\,(6x)$

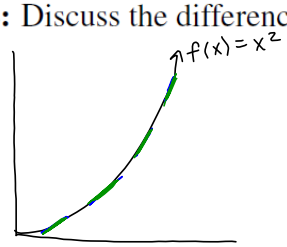
$= 3x + 6x\ln x$ ← 1st deriv.

$y'' = 3 + 6\cancel{x}\cdot\frac{1}{\cancel{x}} + (\ln x)\cdot 6$

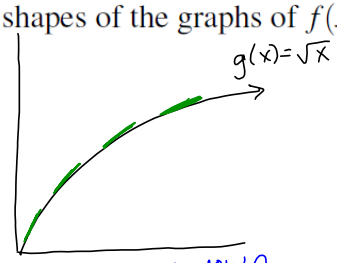
$= 3 + 6 + 6\ln x$

$= 9 + 6\ln x$ ← 2nd deriv.

Example 2: Discuss the difference in shapes of the graphs of $f(x) = x^2$ and $g(x) = \sqrt{x}$ on $[0, \infty)$



concave up
increasing at an increasing rate
slopes are getting larger
f'(x) is increasing



concave down
increasing at a decreasing rate
slopes are getting smaller
f'(x) is decreasing

Definition: The graph of a function f is **concave upward** on the interval (a,b) if f'(x) is increasing on (a,b) and is **concave downward** on the interval (a,b) if f'(x) is decreasing on (a,b).

Example 3: Given $f(x) = -x^3 + 2x^2 - 3x + 9$ determine the intervals where $f'(x)$ is increasing and decreasing.

$f'(x) = -3x^2 + 4x - 3$

$f''(x) = -6x + 4$ ← create a sign chart for the derivative of $f'(x)$

$f''(x) = 0$

$-6x + 4 = 0$

$\frac{4}{6} = \frac{6x}{6}$

$2/3 = x$

$f(x)$	$\cup$		$\cap$
$f'(x)$	$\nearrow$		$\searrow$
$f''(x)$	$+$		$-$
	$x=0$	$2/3$	$x=1$

$f'(x)$ is increasing on $(-\infty, 2/3)$

$f'(x)$ is decreasing on $(2/3, \infty)$

Test for Concavity: For a function whose second derivative exists on an open interval (a,b):

1. If $f''(x) > 0$ (f' is increasing) for all x on (a,b), then $f(x)$ is concave up on (a,b).
2. If $f''(x) < 0$ (f' is decreasing) for all x on (a,b), then $f(x)$ is concave down on (a,b).

Definition: An **inflection point** is a point on the graph of the function where the concavity changes.

Locating Inflection Points:

1. Determine the values of x where the second derivative is zero or where the second derivative is undefined.
2. Place these values on a number line and create a sign chart for the second derivative.
3. The point is an inflection point if the second derivative changes sign AND if the x-value is in the domain of $f(x)$.

Example 4: Determine the intervals where the functions below are concave up and concave down and locate any inflection points.

a) $f(x) = x^4 - 6x^2$ Domain: $\mathbb{R}$

*create a sign chart for 2nd derivative

$f'(x) = 4x^3 - 12x$

$f''(x) = 12x^2 - 12$

$f''(x) = 0$

$12x^2 - 12 = 0$

$12(x^2 - 1) = 0$

$12(x+1)(x-1) = 0$

$x+1=0 \quad x-1=0$

$x=-1 \quad x=1$

$f(x)$	$\cup$		$\cap$		$\cup$
$f''(x)$	$+$		$-$		$+$
	$x=-2$	-1 IP	$x=0$	IP 1	$x=2$

Concave Up $(-\infty,-1) \cup (1,\infty)$

Concave Down $(-1,1)$

Inflection Points $(-1,-5)$ & $(1,-5)$

$f(-1)$ $f(1)$

b) $f(x) = \ln(x^2 + 6x + 13)$ Domain: $\mathbb{R}$

always positive

$f'(x) = \frac{1}{x^2+6x+13} \cdot (2x+6) = \frac{2x+6}{x^2+6x+13}$

$f''(x) = \frac{(x^2+6x+13)(2) - (2x+6)(2x+6)}{(x^2+6x+13)^2}$

$= \frac{2x^2+12x+26 - (4x^2+24x+36)}{(x^2+6x+13)^2}$

$= \frac{2x^2+12x+26-4x^2-24x-36}{(x^2+6x+13)^2} = \frac{-2x^2-12x-10}{(x^2+6x+13)^2}$

$= \frac{-2(x^2+6x+5)}{(x^2+6x+13)^2}$ always positive

No value of x that makes $f''(x)$ undefined

$f''(x) = 0$ when

$-2(x^2+6x+5) = 0$

$-2(x+1)(x+5) = 0$

$x+1=0 \quad x+5=0$

$x=-1 \quad x=-5$

$f(x)$	$\cap$		$\cup$		$\cap$	
$f''(x)$	$-$		$+$		$-$	
	$x=-6$	-5 IP	$x=-2$	-1 IP	$x=0$	

Concave Down $(-\infty,-5) \cup (-1,\infty)$

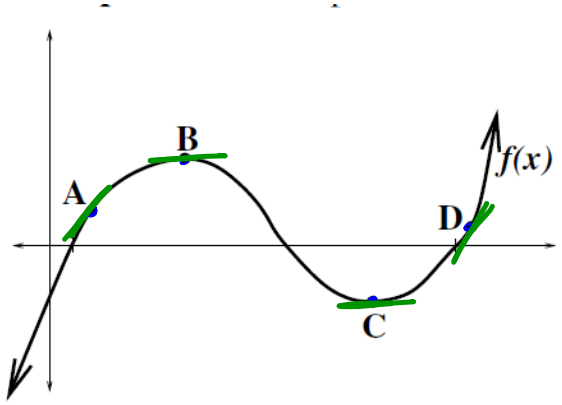
Concave Up $(-5,-1)$

Inflection Points:

$(-5, \ln(8))$ $(-1, \ln(8))$

$f(-5)$ $f(-1)$

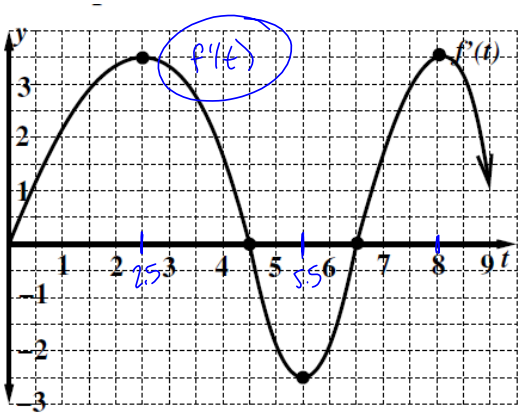
Example 5: At each point below on the graph of $f(x)$ determine whether f, f', and f'' is positive, negative, or zero.



	f	f'	f''
A	+	+	−
B	+	0	−
C	−	0	+
D	+	+	+

Discuss your answer to this problem

Example 6: Given that the domain of $f(t)$ is $[0,\infty]$ and the graph below is of $f'(t)$, determine the following:



a) the interval(s) on which $f(t)$ is concave up.

$(0, 2.5) \cup (5.5, 8)$

b) the interval(s) on which $f(t)$ is concave down.

$(2.5, 5.5) \cup (8, \infty)$

$f(t)$	∪	∩	∪	∩
$f''(t)$	+	−	+	−
$f'(t)$	↗	↘	↗	↘

0 2.5 5.5 8

c) the t-value of the inflection points of $f(t)$.

$t = 2.5, 5.5, 8$

Second Derivative Test for Local Extrema: Let c be a critical value for $f(x)$. ($f'(c)=0$ or $f'(c)$ is undefined and c is in the domain of $f(x)$)

$f'(c)$	$f''(c)$	Concavity?	$f(c)$ is ...
0	+	$\cup$	local min
0	-	$\cap$	local max
0	0	—	test does not apply

Domain: $\mathbb{R}$

Example 7: Determine where the local extrema of $f(x) = -x^3 + 12x - 5$ occur (and classify) using the Second Derivative Test for Local Extrema.

$f'(x) = -3x^2+12$

$f''(x) = -6x$

$f'(x)=0$

$-3x^2+12=0$

$-3(x^2-4)=0$

$-3(x+2)(x-2)=0$

$x+2=0 \quad x-2=0$

$x=-2 \quad x=2$

critical values

$x=-2$

$f''(-2) = -6(-2)$

$= 12$

$\cup$

A local min occurs at $x=-2$

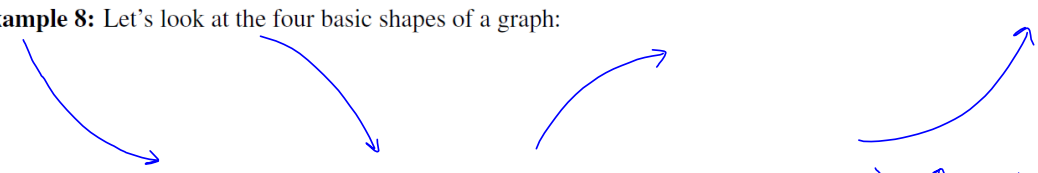
$x=2$

$f''(2) = -6(2) = -12$

$\cap$

A local max occurs at $x=2$.

Example 8: Let's look at the four basic shapes of a graph:



$f(x) \searrow \cup$

$f'(x) < 0$

$f''(x) > 0$

$f(x) \searrow \cap$

$f'(x) < 0$

$f''(x) < 0$

$f(x) \nearrow \cap$

$f'(x) > 0$

$f''(x) < 0$

$f(x) \nearrow \cup$

$f'(x) > 0$

$f''(x) > 0$

Section 5.2 Highly Suggesetd Homework Problems: 1-35(odd), 41-45(odd), 57, 63, 71, 73